

Technical Specification: Unified v2 Pruning

This document provides the mathematical foundations and implementation details for the Unified v2 Pruning method. It addresses the limitations of standard pruning metrics (Magnitude, WANDA, SparseGPT) on noisy time-series data.

1. Error-Weighted Gram Matrix Accumulation

Standard pruning often calculates the Hessian ($X^T X$) using unweighted inputs. Unified v2 modifies this by weighting input windows based on the model's error.

A. Weight Calculation

For each calibration window i , we compute the Mean Squared Error (MSE) of the dense model's prediction against the target.

Let $e_i = \text{MSE}(\text{pred}_i, \text{target}_i)$.

B. Gram Matrix Update

The Gram matrix G (approx Hessian) for each group of 4 inputs is computed as a weighted sum of outer products:

2. Adaptive Unified Scoring

We blend two metrics: Gram-based scoring (standard) and Spectral Quality scoring (noise-robust).

Activation Energy via FFT:

$$E_{\text{signal}} = E_{\text{trend}} + E_{\text{season}}$$

$$E_{\text{noise}} = \text{Energy in high-frequency band (>70\% Nyquist)}$$

$$\text{Noise Fraction (N_frac)} = E_{\text{noise}} / (E_{\text{signal}} + E_{\text{noise}})$$

B. Blending Factor (Alpha) and Unified Score

$$\text{Alpha} = \text{clamp}((\text{N_frac} - 0.15) / 0.15, 0, 1)$$

$$\text{Unified_Score} = \text{Alpha} * Z(\text{Score_spectral}) + (1 - \text{Alpha}) * Z(\text{Score_ratio})$$

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where $Z()$ is Z-normalization across the layer.

3. Auto-Ridge Regularization

To prevent the refit step (Optimal Brain Surgeon) from overfitting to noise, we adapt the regularization strength (λ).

Effective
layer
where
Score

4. Performance Summary

Unified v2 achieves superior results across all ETT datasets (2:4 Sparsity).

Dataset	Unified v2 (MSE)	SparseGPT	Magnitude	WANDA
ETTh1	-0.14 (Best)	+0.03	+0.02	+0.49
ETTh2	-0.18 (Best)	+3.17	+4.65	+6.39
ETTm1	-0.31 (Best)	-0.26	-0.06	+0.44
ETTm2	+0.61 (Best)	+2.47	+5.49	+5.16